Fiscal Year:	FY 2020	**Task Last Updated:**	FY 06/29/2020
PI Name:	Pinsky, Lawrence Ph.D.		
Project Title:	Feasibility Study for Potential Radiation Analog		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	TRISH--TRISH		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Houston	**State:**	TX
Zip Code:	77204-5005	**Congressional District:**	18
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	TRISH--Focused Investigations
Start Date:	01/01/2019	**End Date:**	12/31/2019
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	TRISH
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX16AO69A-FIP0010		
Performance Goal No.:			
Performance Goal Text:			
	Focused Investigation Project Astronaut time on the International Space Station (ISS) is a scarce commodity, and as such real-time evaluations of short-term effects due to radiation exposures such as are experienced within the ISS are difficult to schedule. Flights at aircraft altitudes in low geomagnetic cutoff (i.e., polar) regions experience a Dose Equivalent rate that is comparable enough to that nominally received by astronauts on the ISS to provide a potential analog laboratory to study possible real-time effects. Furthermore, compact instrumentation based on the Timepix technology similar to that deployed on the ISS can be used to accurately assess the radiation environment within the aircraft in real- time. Operational Process; Communications plan Periodic reports will be submitted. A final report will be submitted summarizing the activities undertaken and		

Task Description:	completed. Telecons and face-to-face meetings will be convened as needed to communicate project status and interim findings; Milestones: 1) Timepix detector assemblies obtained: December 1, 2018, 2) Timepix detector assemblies calibrated and incorporated in flight units: February 1, 2019, 3) Initial flight tests completed: June 1, 2019, and 4) Report on all items: September 30, 2019; Compliance with Federal Regulations: All federal regulations will be adhered to, and no explicit human subjects data will be acquired, so no Institutional Review Board (IRB) approvals needed; Follow up: The follow-up will consist of participation in the drafting and submission of the subsequent proposal to the NASA Human Research Program. Timelines: Timepix-based Detector Flight Units available for use: February 1, 2019; Initial Flight Tests completed: June 1, 2019; Final analysis and report complete: September 30, 2019.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The project aim is to evaluate the administrative and technical feasibility for using polar route flights by aircraft as an analog laboratory for evaluation of Dose-Equivalent-Rate effects in radiation environments that are similar to those typically encountered during the majority of the time onboard the International Space Station (ISS). The result of the project and demonstration was to conclude that it would be feasible.
Task Progress:	The goal of the study was to investigate the feasibility of utilizing high polar latitude flights on routine commercial airline routes as well as potential dedicated NASA and/or military flights as a radiation exposure facility. The high cost of engaging astronauts on the International Space Station (ISS) in evaluating the impact of exposure to high background Dose Equivalent Rates led us to ask the question regarding the feasibility to provide such an opportunity to investigators of using polar analog flight opportunities at far less cost than doing so on the ISS. These polar flights also offer a unique laboratory where humans can be exposed to transient (5 to 10 hour) sessions in a radiation environment that has comparable Dose-Equivalent Rates that are typical of the exposure of astronauts to the background radiation environment on a significant portion of the ISS's orbit. One caveat at the outset should be made clear--while the Dose Equivalent Rates are comparable to those portions of the ISS's orbit, the actual constituents of the radiation field are quite different. Our conclusion is that the development and management of such a facility is indeed feasible. The next steps would involve identifying potential users, and formalizing the operational details including not only making use of the repertoire of computer code predictions, but of actively measuring the environment in situ with compact hardware based on Hybrid Radiation Pixel detectors like the Timepix devices from the Medipix Collaborations at the European Laboratory for Particle Physics (CERN) in Geneva, Switzerland. In conjunction with the instrument deployment demonstrations on test flights, we also developed our own (FLUKA-Based) Monte Carlo simulation of our detectors inside an aircraft cabin when exposed at polar flight altitudes. We will submit a separate detailed proposal to continue to the next step.
Bibliography Type:	Description: (Last Updated:)